

# KARL HILL

AEROSPACE & NATIONAL SECURITY SOFTWARE  
ENGINEERING SYSTEMS · TECHNICAL LEADERSHIP

Washington, DC · (202) 599-1442 · karlhillx@gmail.com

linkedin.com/in/khill · github.com/karlhillx · karlhill.com

## SUMMARY

Staff aerospace software engineer with 20+ years building and delivering production software across national security, aerospace, NASA, and enterprise environments. Hands-on in the code and responsible for the engineering system around it—leading software delivery, raising engineering standards, mentoring engineers, resolving cross-team technical problems, and turning mission needs into reliable software.

## PROFESSIONAL EXPERIENCE

### Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs — National Security · Chantilly, VA

- Lead day-to-day engineering and software delivery for a ~10-engineer team across ~20 repositories and multiple environments: priorities, sprint execution, dependencies, and release readiness.
- Coordinate engineering work across the team and vendors—working their sprint cadences, creating and assigning tickets, implementing and delegating work, and clearing dependencies.
- Establish engineering guardrails in CI/CD, automated testing, code quality, security scanning, repository standards, pull requests, and release controls.
- Anticipate technical, integration, and delivery risks; convene the right engineers and stakeholders and drive issues to resolution before they become blockers.
- Design, implement, and review mission software and shared systems—including broker-independent messaging adapters and post-processing pipeline work—while guiding technical approach across services.
- Onboarded and coached six engineers, reinforcing technical expectations, ownership, and consistent engineering practice.

### Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Led design and development of an AWS-based flood-mapping system, automating processing and distribution of satellite-derived flood products.
- Modernized LAADS DAAC web systems, including Find Data search and ordering and near-real-time access, with GitLab CI/CD and Kubernetes delivery.
- Modernized NASA Earth Observatory's web platform, improving performance, user experience, and search discoverability for an audience of approximately 1.5 million monthly visitors during that work.
- Automated a content-registry workflow, reducing manual dataset-registration work by roughly 60% and improving researcher access.
- Built a Ceph-based file and metadata platform for virtual directory mapping and discovery of large scientific datasets.

## DETAILS

Washington, DC

(202) 599-1442

karlhillx@gmail.com

## LINKS

### LinkedIn

linkedin.com/in/khill

### GitHub

github.com/karlhillx

### Website

karlhill.com

## AREAS OF EXPERTISE

- Software Engineering & Technical Leadership
- Mission Software & Distributed Systems
- Engineering Systems & Developer Experience
- CI/CD, Testing & Release Engineering
- Systems Integration & Cross-Team Delivery
- Mentoring, Agile Delivery & Engineering Standards

---

**PROFESSIONAL EXPERIENCE**

---

**Senior Software Engineer**

JAN 2016 — DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel case-management platform connecting counseling workflows, documentation, and billing.
- Built CRM enhancements for customer lifecycle workflows and reporting, and tightened maintenance and security processes with operations.

**Senior Software Engineer**

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

**Earlier Engineering Experience**

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- Built Java and SQL Server services for Finium, a multi-tenant managed-security platform, and contributed to shared testing and code-quality practices.

---

**AREAS OF EXPERTISE**

---

Software Engineering & Technical Leadership · Mission Software & Distributed Systems · Engineering Systems & Developer Experience · CI/CD, Testing & Release Engineering · Systems Integration & Cross-Team Delivery · Mentoring, Agile Delivery & Engineering Standards

---

**EDUCATION**

---

**Computer Science coursework**, University of Maryland, Baltimore County

**Associate of Arts, General Studies — Computer Science Curriculum**, Howard Community College

**Project Management studies**, Rutgers University

---

**CERTIFICATIONS**

---

**SAFe Agilist**, Scaled Agile (in progress)

**Professional Scrum Master II**, Scrum.org

**Professional Scrum Product Owner I**, Scrum.org

---

**DEVELOPER TOOLING**

---

**bb-run** — Run Bitbucket Pipelines locally.

**testrisk** — Rank the highest-value Python test gaps from coverage, AST, and git.

---

**TECHNICAL EXPERTISE**

---

**Languages:** Python, TypeScript, Java, PHP, Perl, Bash

**Cloud & Containers:** AWS, Docker, Kubernetes, Helm

**CI/CD & Developer Tooling:** Git, Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation, security scanning

**Python Tooling:** uv, Ruff, pytest, ty, pre-commit

**Integration:** REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems

**Data:** PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis